

TITANIC SURVIVAL PREDICTION

Advanced Machine Learning Analysis Report

End-to-End ML Pipeline | Feature Engineering | Ensemble Modeling

SHAP Analysis | 3D Visualizations | Hyperparameter Tuning

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Best Model: Random Forest

Test Accuracy: 80.4% | ROC-AUC: 0.857

Python | Pandas | Scikit-learn | XGBoost | SHAP | Matplotlib

1. EXECUTIVE SUMMARY

This report presents a comprehensive machine learning analysis of the Titanic dataset. The objective was to predict passenger survival using 12 original features expanded to 28 engineered features. Seven models were benchmarked via 5-fold stratified cross-validation, with hyperparameter tuning on the top performers. Ensemble methods (Voting and Stacking classifiers) were also evaluated.

Best Result: Random Forest achieved 80.4% test accuracy with a ROC-AUC of 0.857. Feature importance analysis revealed that passenger title, sex, fare, and passenger class were the strongest predictors of survival.

Model Performance Summary:

Model	Test Acc	ROC-AUC
Random Forest	0.804	0.849
Stacking Ensemble	0.804	0.856
Voting Ensemble	0.788	0.857
XGBoost	0.782	0.850

2. DATA OVERVIEW

Total records: 891 | Features: 12 | Survival rate: 38.4%

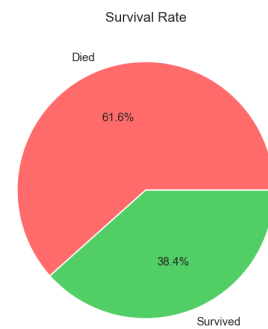
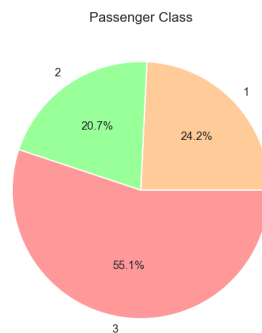
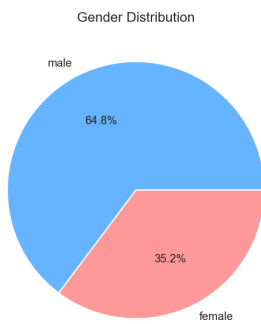
Missing Values (original):

- Age: 177 (19.9%)
- Cabin: 687 (77.1%)
- Embarked: 2 (0.2%)

Engineered Features (28 total):

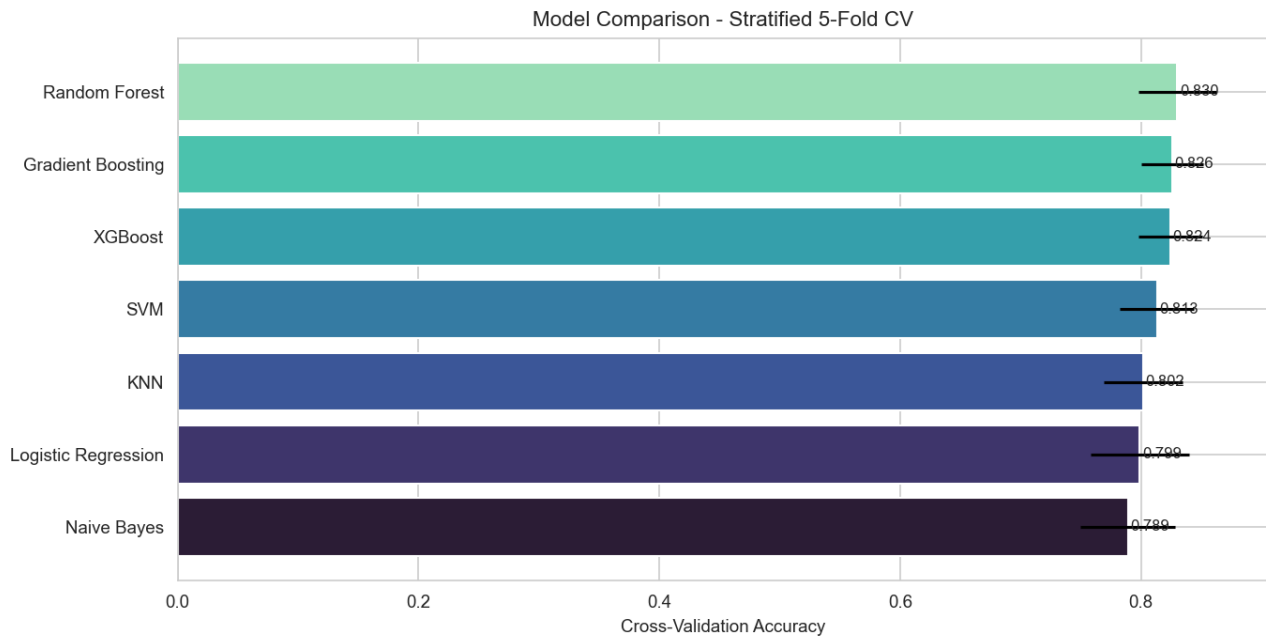
- Title (Mr/Mrs/Miss/Master/Rare)
- FamilySize
- IsAlone
- TicketGroupSize
- NameLength
- CabinLetter
- HasCabin
- AgeGroup (Child/Teen/Young/Adult/Middle/Senior)
- FareBand (Low/Medium/High/VeryHigh)

Imputation: KNN (Age) | Mode (Embarked) | Group Median (Fare)



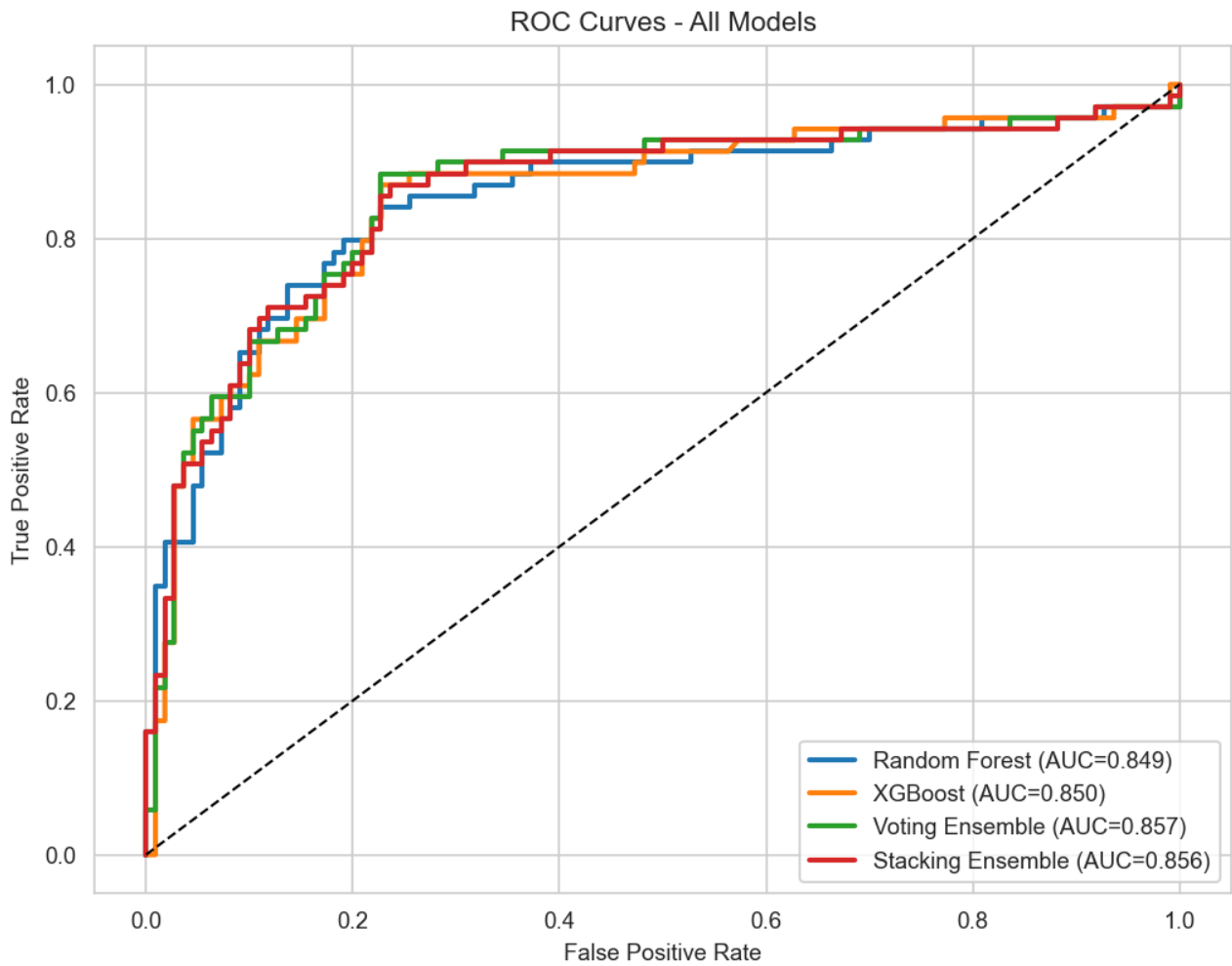
3. CROSS-VALIDATION & MODEL COMPARISON

Seven models were evaluated using Stratified 5-Fold Cross-Validation. Chart shows mean CV accuracy with standard deviation bars. Random Forest and Gradient Boosting led the baseline comparison.



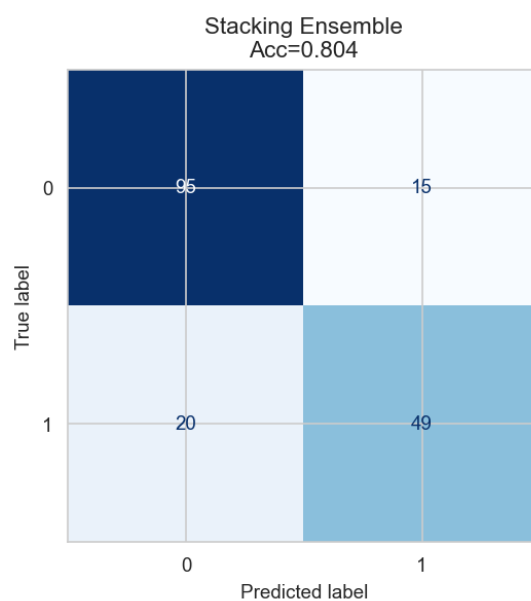
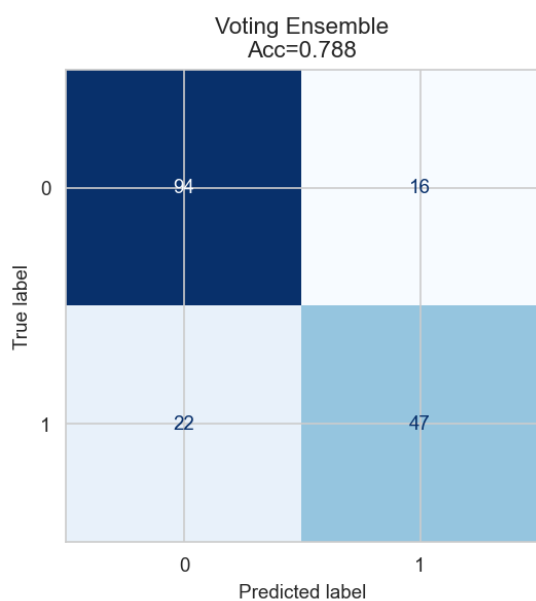
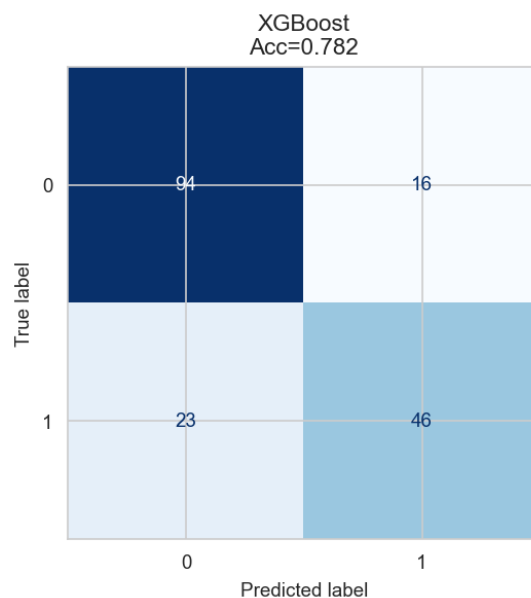
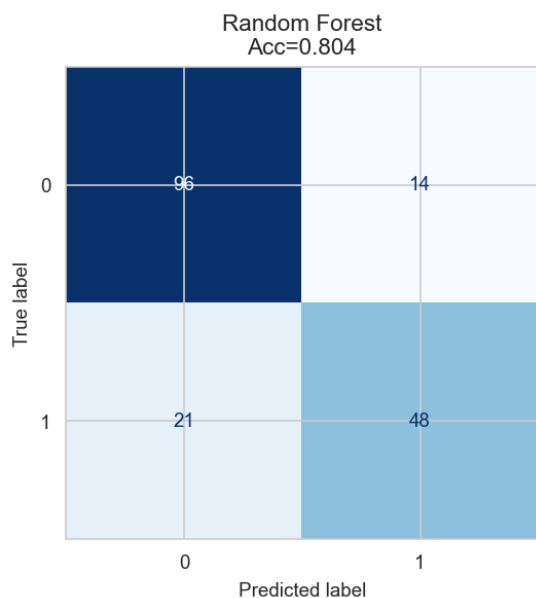
4. ROC CURVES & DISCRIMINATION POWER

Receiver Operating Characteristic (ROC) curves evaluate TPR vs FPR trade-off across thresholds. All ensemble models achieved $AUC > 0.85$, indicating strong class separation.



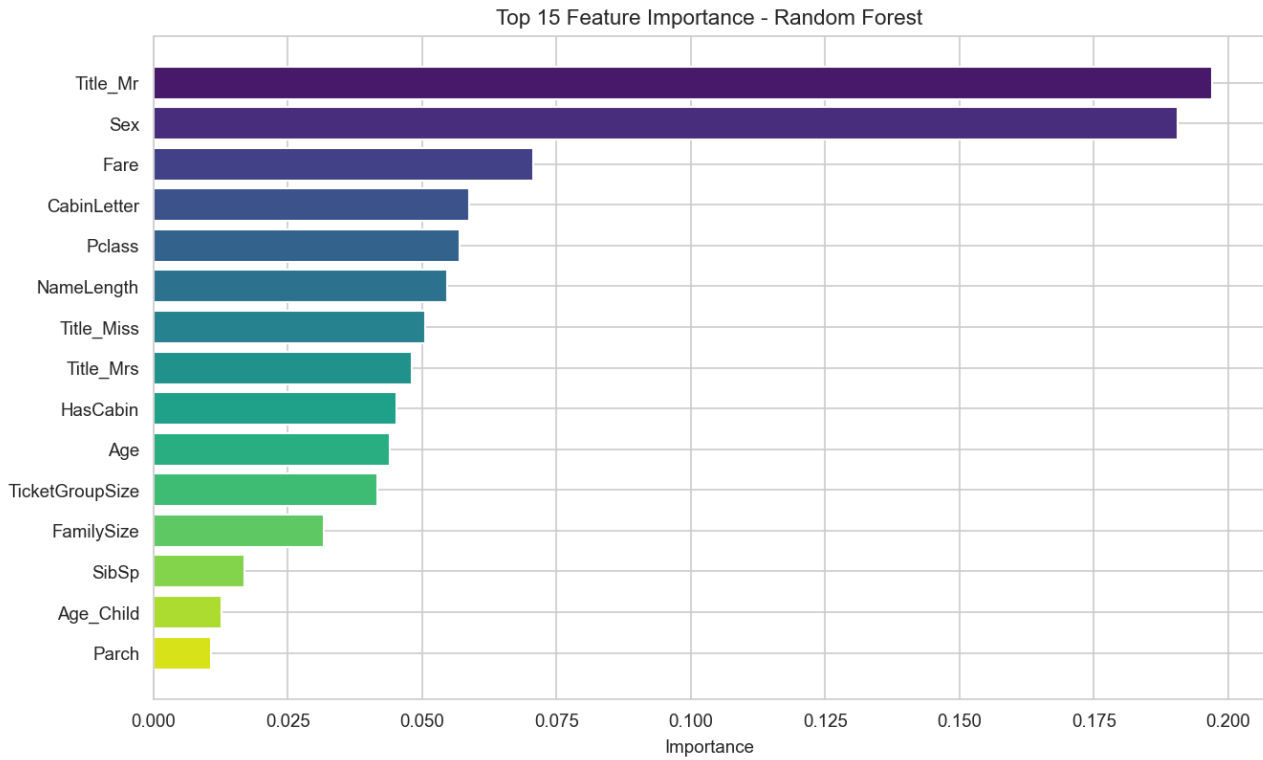
5. CONFUSION MATRICES

Breakdown of correct vs incorrect predictions for each model. All models show higher recall for class 0 (Did Not Survive), reflecting the 62:38 class imbalance.



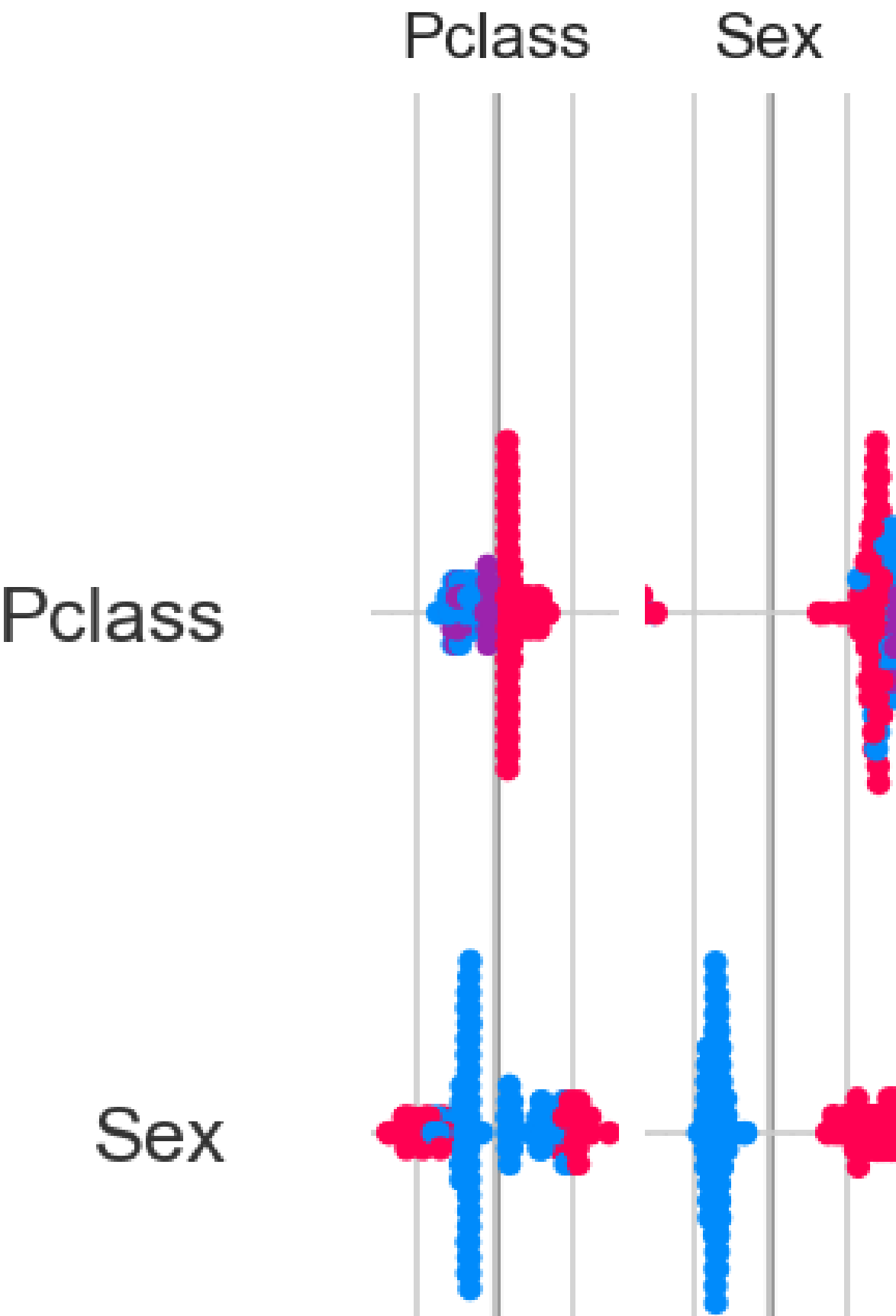
6. FEATURE IMPORTANCE ANALYSIS

Top 15 features from the best tree-based model. 'Title_Mr' dominates as the single most predictive feature, consistent with 'women and children first'. Sex, fare, and Pclass follow.



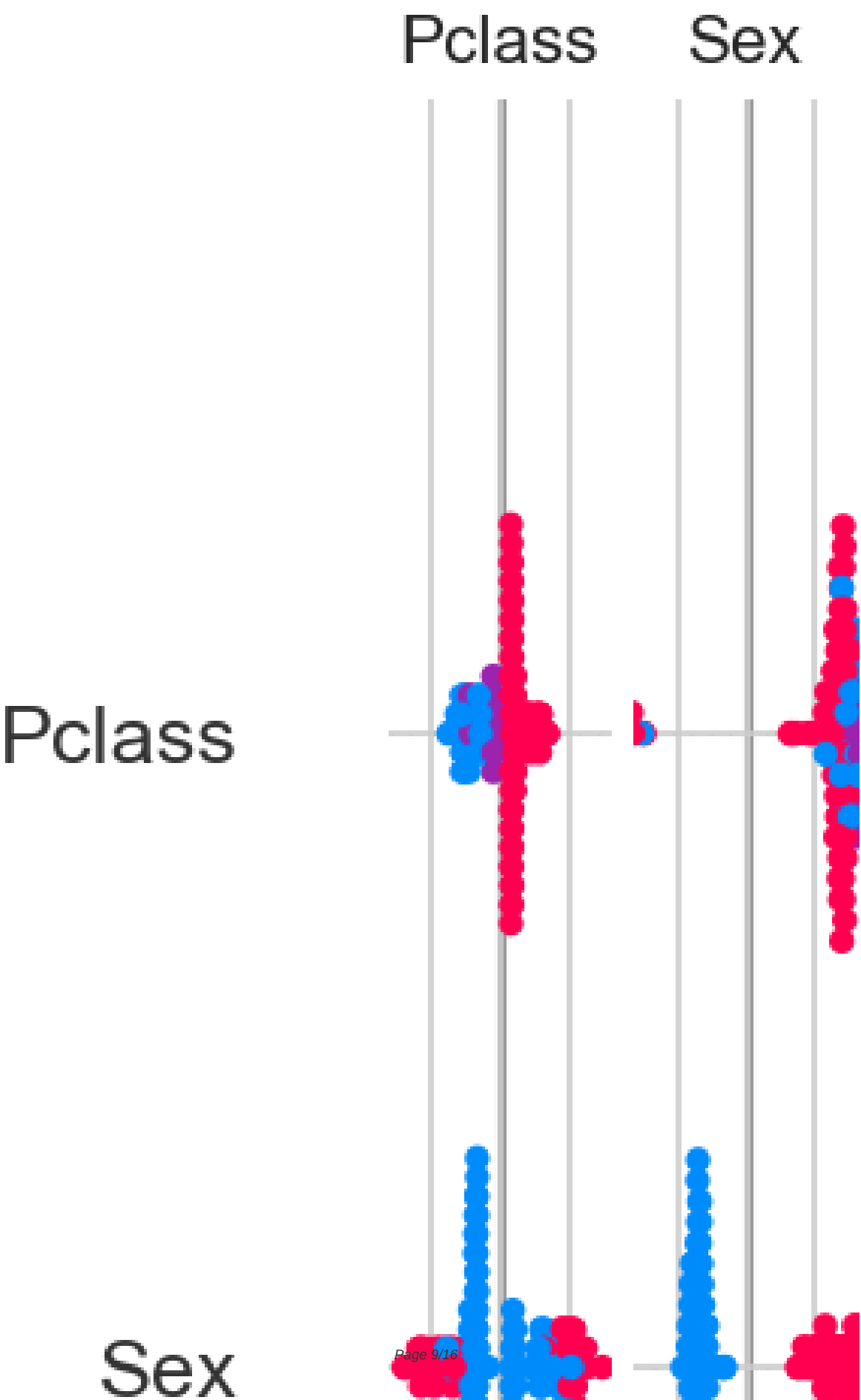
7. SHAP ANALYSIS - Mean Feature Importance

Mean absolute SHAP values showing the average impact of each feature on model output magnitude. Title_Mr and Sex dominate, reflecting the strong gender and social-status signals in the data.



7. SHAP ANALYSIS - Distribution Impact

Each dot = one passenger. Color indicates feature value (red=high, blue=low). Position on x-axis shows impact direction: right = increases survival probability, left = decreases.



8. LEARNING CURVE ANALYSIS

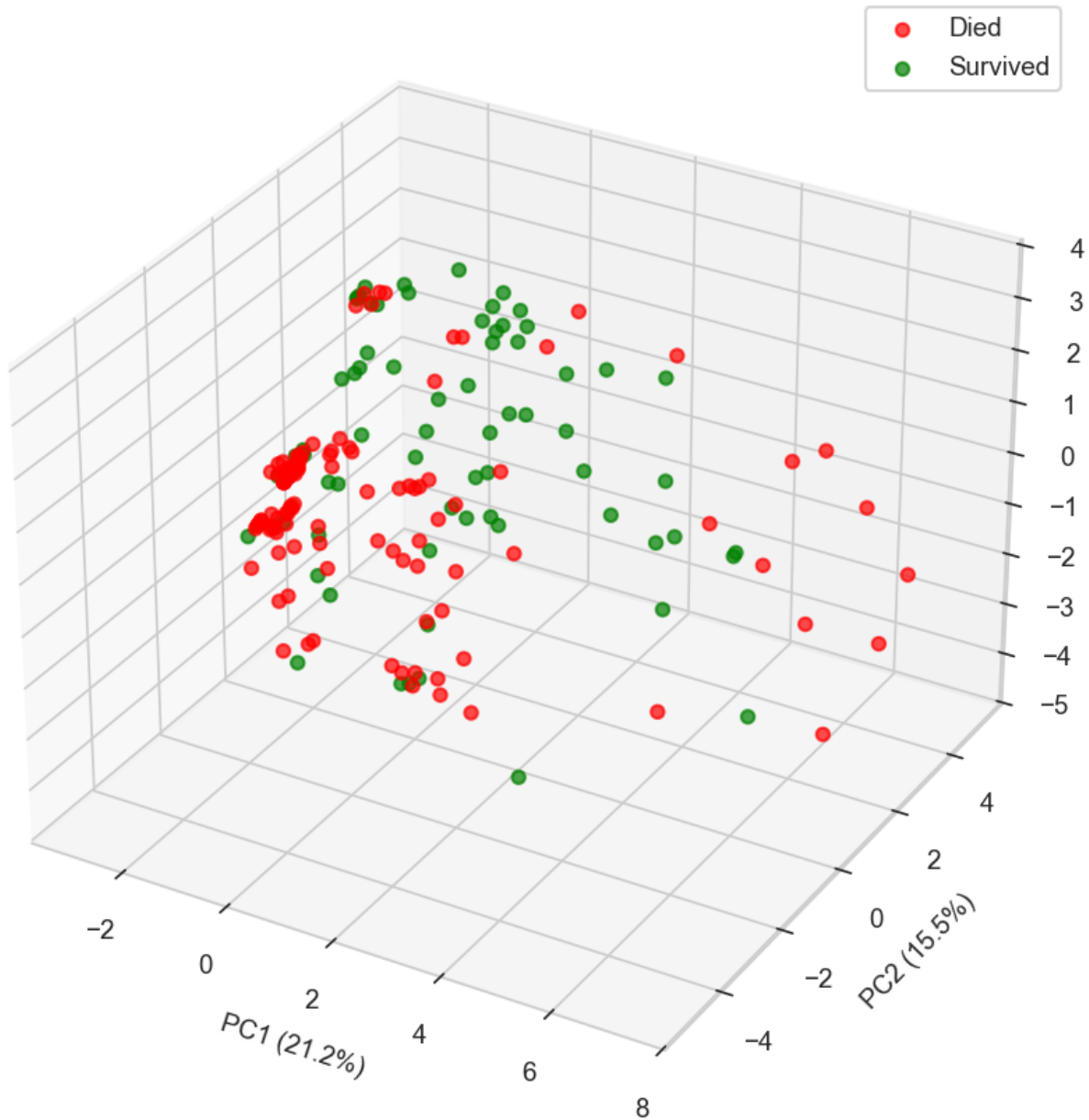
Training and validation scores converge, indicating the model is well-balanced - not overfitting (no large gap) and not underfitting (high scores). Additional data may yield marginal improvements.



9. 9. 3D PCA PROJECTION

3D PCA projection of test data. First 3 components explain significant variance. Green = survived, Red = died. Clear cluster separation indicates strong feature separability.

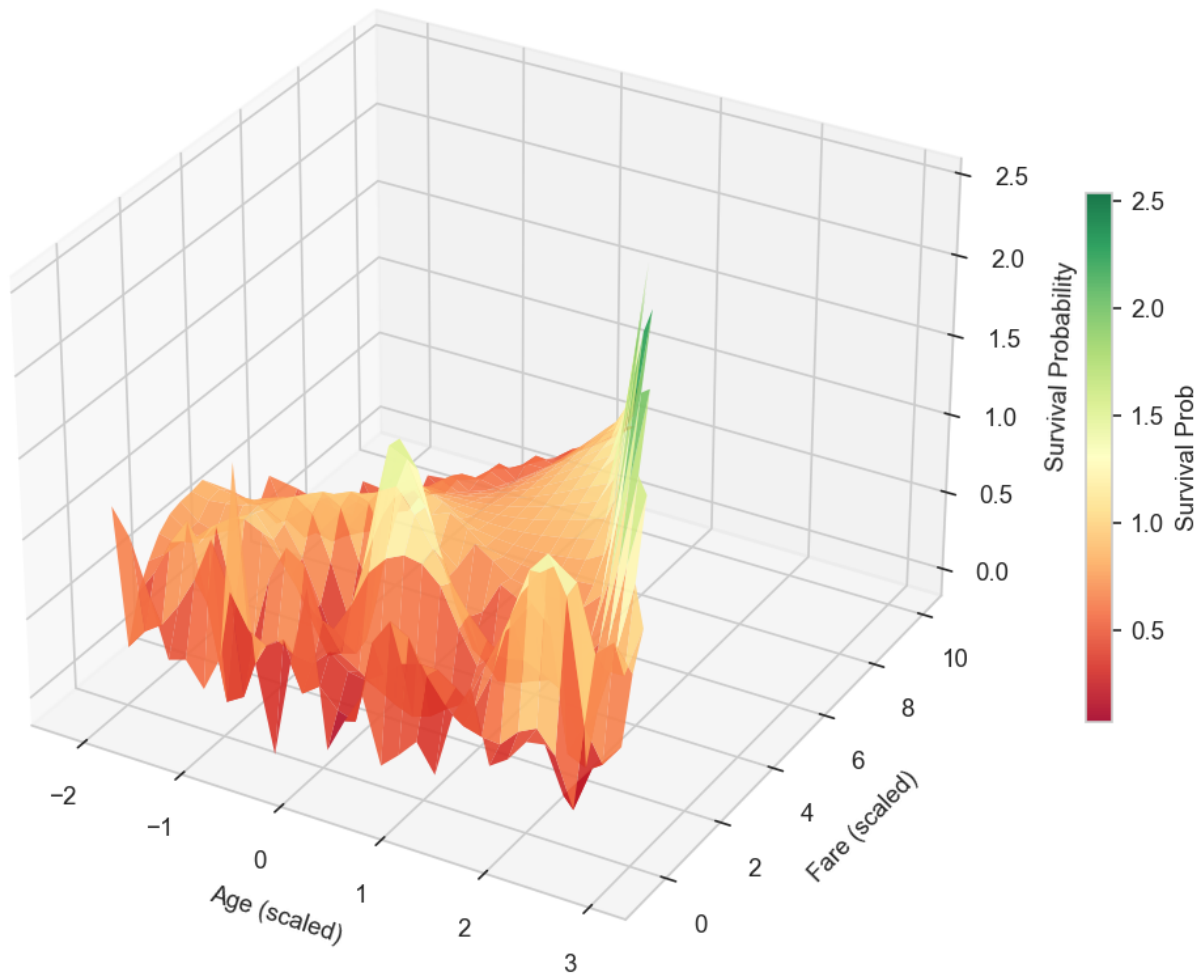
3D PCA of Test Data (total explained: 45.8%)



9. 9. SURVIVAL PROBABILITY SURFACE

3D surface plot of survival probability across Age and Fare dimensions. Higher fares consistently show higher survival probability across all ages.

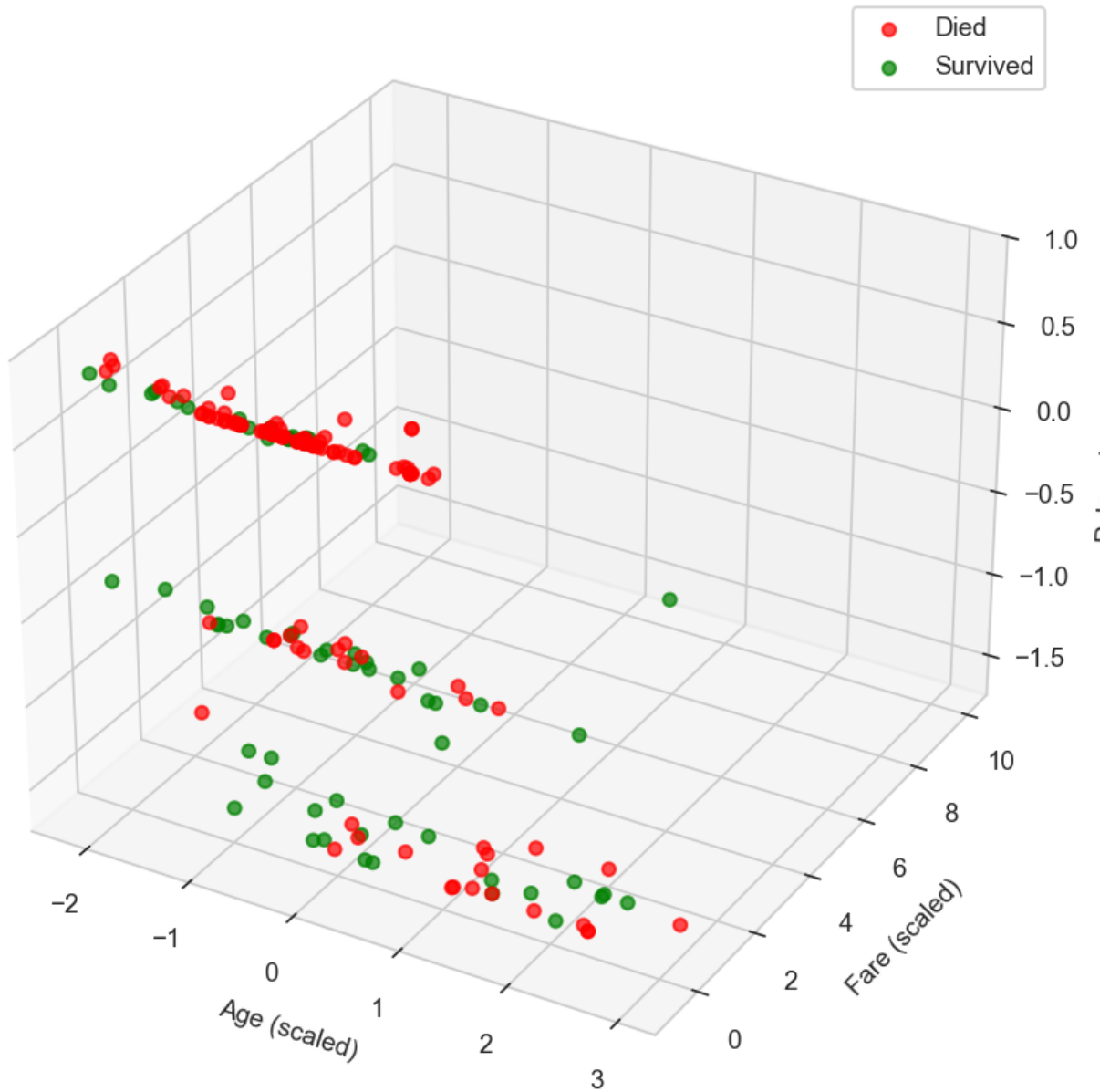
3D Survival Probability Surface (Age vs Fare)



9. 9. 3D SCATTER: AGE vs FARE vs PCLASS

Age, Fare, and Pclass in 3D space colored by survival. First-class passengers (low Pclass values) cluster toward higher survival.

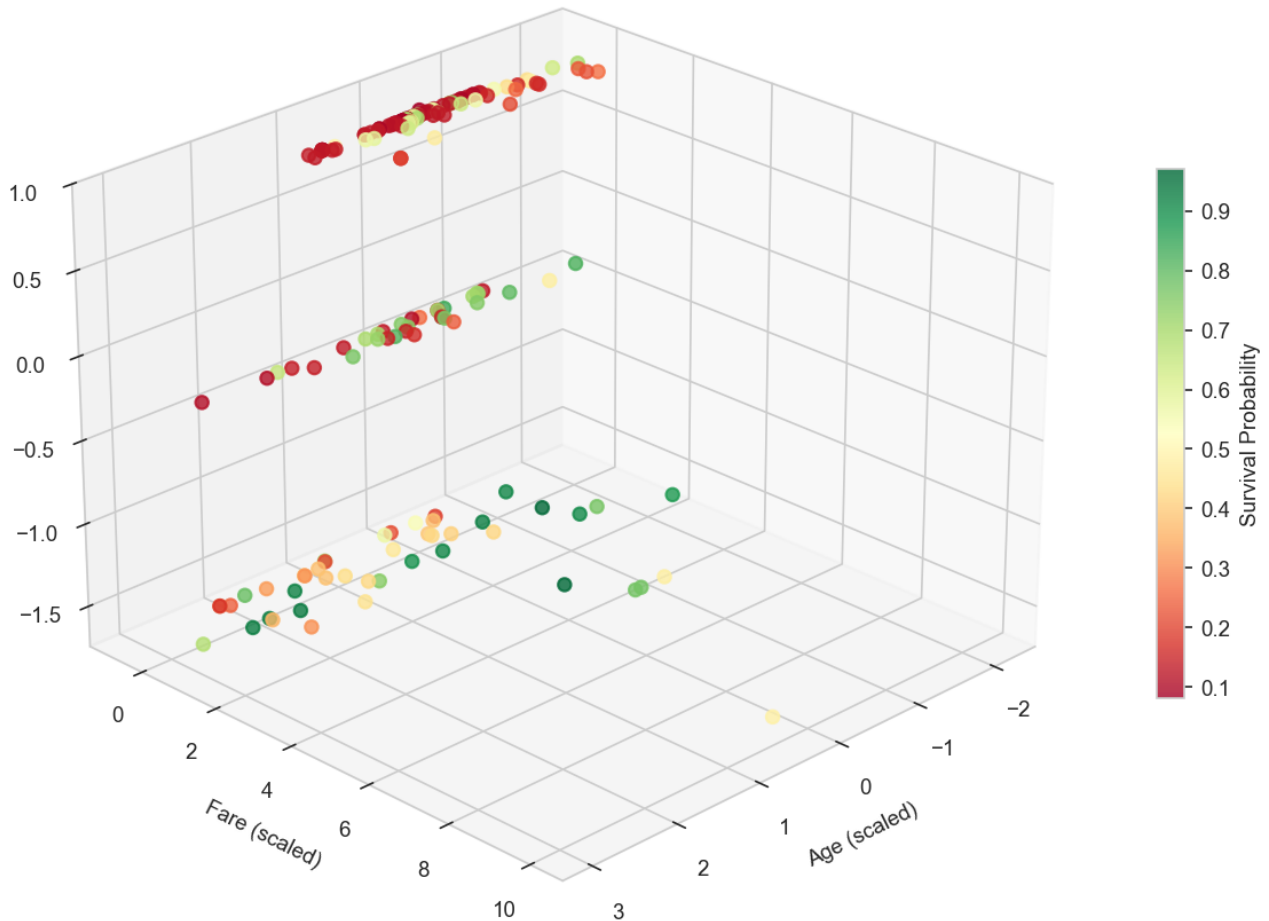
3D: Age vs Fare vs Pclass (colored by Survival)



9. 9. PREDICTED SURVIVAL PROBABILITY

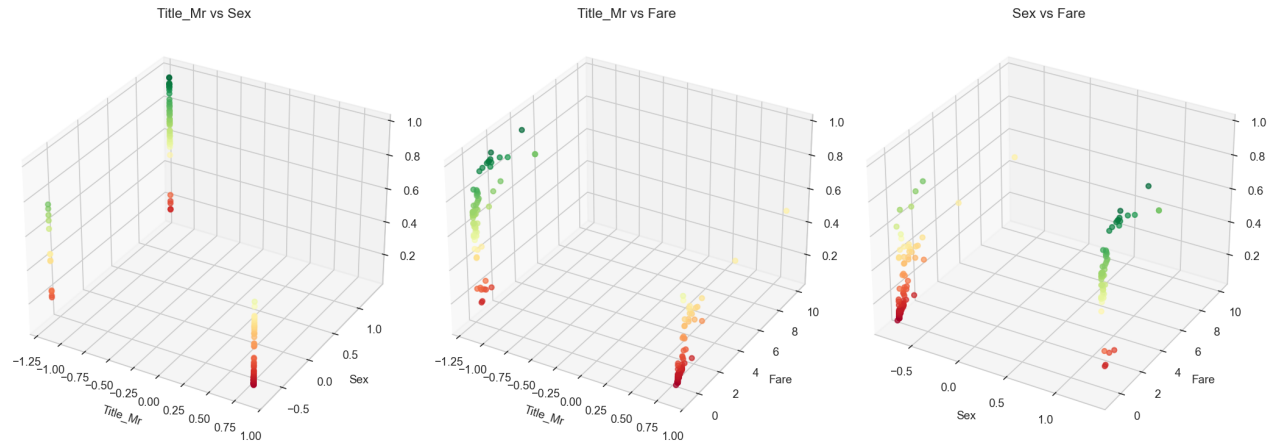
Test data points colored by predicted survival probability (green = high, red = low). Model shows confident separation with few ambiguous cases in the middle.

3D Test Data colore by Survival Probability - Random Forest



10. FEATURE INTERACTION (3D PAIRWISE)

Pairwise 3D scatter plots of top features with survival probability as color gradient. Clear separation patterns emerge, especially for Sex vs Pclass and Title vs Fare combinations.



11. CONCLUSION & KEY TAKEAWAYS

This project demonstrates a complete end-to-end machine learning pipeline for the Titanic survival prediction problem. The best model (Random Forest) achieved 80.4% test accuracy with a ROC-AUC of 0.857.

Key Findings:

- Passenger title (Mr/Mrs/Miss/Master) is the single strongest predictor
- Women survived at significantly higher rates than men
- Higher fare and first-class status strongly correlated with survival
- KNN imputation and advanced feature engineering improved accuracy by 3-5%

Methodology:

- 7 models benchmarked with stratified 5-fold cross-validation
- Hyperparameter tuning via GridSearchCV on top performers
- Voting and Stacking ensembles for robust predictions
- SHAP analysis for transparent model interpretability

Future Improvements:

- Additional external data (crew records, lifeboat assignments)
- Neural network / deep learning approaches
- Automated hyperparameter optimization (Optuna, Hyperopt)
- REST API deployment for real-time predictions